

1. FILE SYSTEM NAVIGATION

Essential commands to locate, inspect, and transition between directories across the Linux filesystem.

`pwd`
Print Working Directory: Displays full absolute path of current location.

`ls [options] [path]`
List Directory Contents:

- `ls -l` : Long listing format (permissions, owner, size, timestamp).
- `ls -a` : Shows all files including hidden files (starting with `.`).
- `ls -la /usr/share/wordlists` : Comprehensive listing of specific path.

`cd [directory]`
Change Directory:

- `cd /var/www/html` : Navigate to explicit directory.
- `cd ~` or `cd` : Return to user's home directory (e.g., `/home/kali`).
- `cd ..` : Move up one directory level.
- `cd -` : Toggle back to previous working directory.

2. FILE & DIRECTORY PERMISSIONS

Linux utilizes a permission model defined across **Owner (User)**, **Group**, and **Others**.

Permission String Anatomy: `-rwxr-xr--`
`-` (File Type) | `rwx` (Owner) | `r-x` (Group) | `r--` (Others)

PERMISSION	SYMBOL	OCTAL VALUE	FILE IMPACT	DIRECTORY IMPACT
Read	<code>r</code>	<code>4</code>	Read file content	List directory contents
Write	<code>w</code>	<code>2</code>	Modify file content	Create/delete files in directory
Execute	<code>x</code>	<code>1</code>	Run binary / script	Enter/cd into directory

Modifying Permissions (`chmod`) & Ownership (`chown`):

```
# Calculate total octal sum per user class (Owner / Group / Others)
chmod 755 script.sh      # Owner: rwx (4+2+1=7), Group: r-x (4+1=5), Others: r-x (4+1=5)
chmod 600 id_rsa         # Read/Write for Owner only (SSH private key standard)
chmod +x payload.py      # Grant execute flag to all classes

# Change ownership user and group
sudo chown kali:kali script.sh      # Set user 'kali' and group 'kali'
sudo chown -R kali:kali /opt/tools   # Apply recursively to directory tree
```

3. PACKAGE MANAGEMENT

Kali Linux builds on Debian, utilizing high-level `apt` and low-level `dpkg`.

APT (Advanced Package Tool) - Repositories:

```
sudo apt update      # Refresh repository package lists
sudo apt upgrade -y   # Upgrade all installed packages to latest versions
sudo apt install nmap -y  # Search repo, download, and install tool
sudo apt remove nmap    # Remove package while preserving configs
sudo apt purge nmap     # Remove package AND configuration files
sudo apt search exploitdb  # Search remote indices for relevant packages
```

DPKG - Local Debian (.deb) Packages:

```
sudo dpkg -i custom_tool.deb # Install local .deb file
sudo dpkg -l                  # List all locally installed Debian packages
sudo dpkg -r toolname         # Remove package installed via dpkg
```

PRO-TIP: Resolving DPKG Dependency Failures

If manual `dpkg -i` installation fails due to missing dependencies, run:
`sudo apt --fix-broken install` to automatically fetch and configure missing prerequisites.

4. NETWORKING COMMANDS

Network interface configuration, connectivity troubleshooting, and active socket analysis.

```
# Network Configuration & Interface Status
ifconfig          # View active interfaces, IP, MAC & netmask (legacy)
ip a              # Modern equivalent: display all interface details
sudo ifconfig eth0 up # Bring eth0 network interface online

# Connectivity & Route Testing
ping -c 4 8.8.8.8 # Send 4 ICMP echo requests to verify connectivity
traceroute target.com # Display hop-by-hop route path to destination

# Socket & Port Inspection
netstat -tuln      # List open ports (TCP/UDP, Listening, Numeric format)
netstat -antp      # List active TCP connections with process PID
ss -tuln           # Modern faster alternative to netstat
```

5. QUICK COMMAND SUMMARY

CATEGORY	COMMAND	PRIMARY USE CASE
Navigation	<code>pwd</code> , <code>ls -la</code> , <code>cd</code>	Find, list, and move across directories.
Permissions	<code>chmod 755</code> , <code>chown user:grp</code>	Grant execution rights and manage ownership.
Packages	<code>apt update && apt install</code>	Keep system patched and install pentesting tools.
Networking	<code>ip a</code> , <code>ping</code> , <code>ss -tuln</code>	Verify local network setup and inspect listening ports.